

definition of the *solution* $y = \phi(x)$ in part (b). Use a graphing utility or a CAS to graph the solution curve for the IVP on this interval.

60. (a) Use a CAS and the concept of level curves to plot representative graphs of members of the family of solutions of the differential equation $\frac{dy}{dx} = \frac{x(1-x)}{y(-2+y)}$. Experiment with different numbers of level curves as well as various rectangular regions in the xy -plane until your result resembles Figure 2.2.6.
- (b) On separate coordinate axes, plot the graph of the implicit solution corresponding to the initial condition $y(0) = \frac{3}{2}$. Use a colored pencil to mark off that segment of the graph that corresponds to the solution curve of a solution ϕ that satisfies the initial

condition. With the aid of a root-finding application of a CAS, determine the approximate largest interval I of definition of the solution ϕ . [Hint: First find the points on the curve in part (a) where the tangent is vertical.]

- (c) Repeat part (b) for the initial condition $y(0) = -2$.

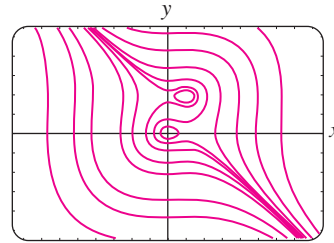


FIGURE 2.2.6 Level curves in Problem 60

2.3 LINEAR EQUATIONS

REVIEW MATERIAL

- Review the definitions of linear DEs in (6) and (7) of Section 1.1

INTRODUCTION We continue our quest for solutions of first-order differential equations by next examining linear equations. Linear differential equations are an especially “friendly” family of differential equations, in that, given a linear equation, whether first order or a higher-order kin, there is always a good possibility that we can find some sort of solution of the equation that we can examine.

A Definition The form of a linear first-order DE was given in (7) of Section 1.1. This form, the case when $n = 1$ in (6) of that section, is reproduced here for convenience.

DEFINITION 2.3.1 Linear Equation

A first-order differential equation of the form

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x), \quad (1)$$

is said to be a linear equation in the variable y .

Standard Form By dividing both sides of (1) by the lead coefficient $a_1(x)$, we obtain a more useful form, the **standard form**, of a linear equation:

$$\frac{dy}{dx} + P(x)y = f(x). \quad (2)$$

We seek a solution of (2) on an interval I for which both coefficient functions P and f are continuous.

Before we examine a general procedure for solving equations of form (2) we note that in some instances (2) can be solved by separation of variables. For example, you should verify that the equations

We match each equation with (2). In the first equation $P(x) = 2x$, $f(x) = 0$ and in the second $P(x) = -1$, $f(x) = 5$.

$$\frac{dy}{dx} + 2xy = 0 \quad \text{and} \quad \frac{dy}{dx} = y + 5$$

are both linear and separable, but that the linear equation

$$\frac{dy}{dx} + y = x$$

is not separable.

Method of Solution The method for solving (2) hinges on a remarkable fact that the *left-hand side* of the equation can be recast into the form of the exact derivative of a product by multiplying the both sides of (2) by a special function $\mu(x)$. It is relatively easy to find the function $\mu(x)$ because we want

$$\frac{d}{dx} [\underbrace{\mu(x)y}_{\text{product}}] = \underbrace{\mu \frac{dy}{dx} + \frac{d\mu}{dx} y}_{\text{product rule}} = \underbrace{\mu \frac{dy}{dx}}_{\text{left hand side of (2) multiplied by } \mu(x)} + \underbrace{\mu P y}_{\text{these must be equal}}.$$

The equality is true provided that

$$\frac{d\mu}{dx} = \mu P.$$

The last equation can be solved by separation of variables. Integrating

$$\frac{d\mu}{\mu} = P dx \quad \text{and solving} \quad \ln|\mu(x)| = \int P(x) dx + c_1$$

See Problem 50
in Exercises 2.3

gives $\mu(x) = c_2 e^{\int P(x) dx}$. Even though there are an infinite choices of $\mu(x)$ (all constant multiples of $e^{\int P(x) dx}$), all produce the same desired result. Hence we can simplify life and choose $c_2 = 1$. The function

$$\mu(x) = e^{\int P(x) dx} \quad (3)$$

is called an **integrating factor** for equation (2).

Here is what we have so far: We multiplied both sides of (2) by (3) and, by construction, the left-hand side is the derivative of a product of the integrating factor and y :

$$e^{\int P(x) dx} \frac{dy}{dx} + P(x) e^{\int P(x) dx} y = e^{\int P(x) dx} f(x)$$

$$\frac{d}{dx} [e^{\int P(x) dx} y] = e^{\int P(x) dx} f(x).$$

Finally, we discover why (3) is called an *integrating factor*. We can integrate both sides of the last equation,

$$e^{\int P(x) dx} y = \int e^{\int P(x) dx} f(x) dx + c$$

and solve for y . The result is a one-parameter family of solutions of (2):

$$y = e^{-\int P(x) dx} \int e^{\int P(x) dx} f(x) dx + c e^{-\int P(x) dx}. \quad (4)$$

We emphasize that you should **not memorize** formula (4). The following procedure should be worked through each time.

SOLVING A LINEAR FIRST-ORDER EQUATION

- (i) Remember to put a linear equation into the standard form (2).
- (ii) From the standard form of the equation identify $P(x)$ and then find the integrating factor $e^{\int P(x)dx}$. No constant need be used in evaluating the indefinite integral $\int P(x)dx$.
- (iii) Multiply the both sides of the standard form equation by the integrating factor. The left-hand side of the resulting equation is automatically the derivative of the product of the integrating factor $e^{\int P(x)dx}$ and y :

$$\frac{d}{dx} [e^{\int P(x)dx} y] = e^{\int P(x)dx} f(x).$$

- (iv) Integrate both sides of the last equation and solve for y .

EXAMPLE 1 Solving a Linear Equation

Solve $\frac{dy}{dx} - 3y = 0$.

SOLUTION This linear equation can be solved by separation of variables. Alternatively, since the differential equation is already in standard form (2), we identify $P(x) = -3$, and so the integrating factor is $e^{\int(-3)dx} = e^{-3x}$. We then multiply the given equation by this factor and recognize that

$$e^{-3x} \frac{dy}{dx} - 3e^{-3x} y = e^{-3x} \cdot 0 \quad \text{is the same as} \quad \frac{d}{dx} [e^{-3x} y] = 0.$$

Integration of the last equation,

$$\int \frac{d}{dx} [e^{-3x} y] dx = \int 0 dx$$

then yields $e^{-3x} y = c$ or $y = ce^{3x}$, $-\infty < x < \infty$. ≡

EXAMPLE 2 Solving a Linear Equation

Solve $\frac{dy}{dx} - 3y = 6$.

SOLUTION This linear equation, like the one in Example 1, is already in standard form with $P(x) = -3$. Thus the integrating factor is again e^{-3x} . This time multiplying the given equation by this factor gives

$$e^{-3x} \frac{dy}{dx} - 3e^{-3x} y = 6e^{-3x} \quad \text{and so} \quad \frac{d}{dx} [e^{-3x} y] = 6e^{-3x}.$$

Integrating the last equation,

$$\int \frac{d}{dx} [e^{-3x} y] dx = 6 \int e^{-3x} dx \quad \text{gives} \quad e^{-3x} y = -6 \left(\frac{e^{-3x}}{3} \right) + c,$$

or $y = -2 + ce^{3x}$, $-\infty < x < \infty$. ≡

When a_1 , a_0 , and g in (1) are constants, the differential equation is autonomous. In Example 2 you can verify from the normal form $dy/dx = 3(y + 2)$ that -2 is a critical point and that it is unstable (a repeller). Thus a solution curve with an

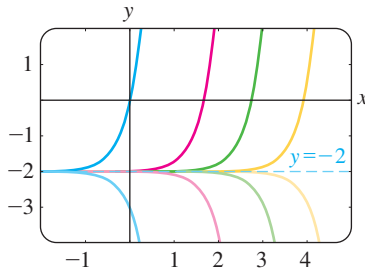


FIGURE 2.3.1 Solution curves of DE in Example 2

initial point either above or below the graph of the equilibrium solution $y = -2$ pushes away from this horizontal line as x increases. Figure 2.3.1, obtained with the aid of a graphing utility, shows the graph of $y = -2$ along with some additional solution curves.

General Solution Suppose again that the functions P and f in (2) are continuous on a common interval I . In the steps leading to (4) we showed that if (2) has a solution on I , then it must be of the form given in (4). Conversely, it is a straightforward exercise in differentiation to verify that any function of the form given in (4) is a solution of the differential equation (2) on I . In other words, (4) is a one-parameter family of solutions of equation (2) and *every solution of (2) defined on I is a member of this family*. Therefore we call (4) the **general solution** of the differential equation on the interval I . (See the *Remarks* at the end of Section 1.1.) Now by writing (2) in the normal form $y' = F(x, y)$, we can identify $F(x, y) = -P(x)y + f(x)$ and $\partial F/\partial y = -P(x)$. From the continuity of P and f on the interval I we see that F and $\partial F/\partial y$ are also continuous on I . With Theorem 1.2.1 as our justification, we conclude that there exists one and only one solution of the initial-value problem

$$\frac{dy}{dx} + P(x)y = f(x), \quad y(x_0) = y_0 \quad (5)$$

defined on *some* interval I_0 containing x_0 . But when x_0 is in I , finding a solution of (5) is just a matter of finding an appropriate value of c in (4)—that is, to each x_0 in I there corresponds a distinct c . In other words, the interval I_0 of existence and uniqueness in Theorem 1.2.1 for the initial-value problem (5) is the entire interval I .

EXAMPLE 3 General Solution

Solve $x \frac{dy}{dx} - 4y = x^6 e^x$.

SOLUTION Dividing by x , the standard form of the given DE is

$$\frac{dy}{dx} - \frac{4}{x}y = x^5 e^x. \quad (6)$$

From this form we identify $P(x) = -4/x$ and $f(x) = x^5 e^x$ and further observe that P and f are continuous on $(0, \infty)$. Hence the integrating factor is

$$\begin{aligned} &\text{we can use } \ln x \text{ instead of } \ln |x| \text{ since } x > 0 \\ &\quad \downarrow \\ &e^{-4 \int dx/x} = e^{-4 \ln x} = e^{\ln x^{-4}} = x^{-4}. \end{aligned}$$

Here we have used the basic identity $b^{\log_b N} = N$, $N > 0$. Now we multiply (6) by x^{-4} and rewrite

$$x^{-4} \frac{dy}{dx} - 4x^{-5}y = xe^x \quad \text{as} \quad \frac{d}{dx}[x^{-4}y] = xe^x.$$

It follows from integration by parts that the general solution defined on the interval $(0, \infty)$ is $x^{-4}y = xe^x - e^x + c$ or $y = x^5 e^x - x^4 e^x + cx^4$. ≡

In case you are wondering why the interval $(0, \infty)$ is important in Example 3, read this paragraph and the paragraph following Example 4. ▶

Except in the case in which the lead coefficient is 1, the recasting of equation (1) into the standard form (2) requires division by $a_1(x)$. Values of x for which $a_1(x) = 0$ are called **singular points** of the equation. Singular points are potentially troublesome. Specifically, in (2), if $P(x)$ (formed by dividing $a_0(x)$ by $a_1(x)$) is discontinuous at a point, the discontinuity may carry over to solutions of the differential equation.

EXAMPLE 4 General Solution

Find the general solution of $(x^2 - 9) \frac{dy}{dx} + xy = 0$.

SOLUTION We write the differential equation in standard form

$$\frac{dy}{dx} + \frac{x}{x^2 - 9}y = 0 \quad (7)$$

and identify $P(x) = x/(x^2 - 9)$. Although P is continuous on $(-\infty, -3)$, $(-3, 3)$, and $(3, \infty)$, we shall solve the equation on the first and third intervals. On these intervals the integrating factor is

$$e^{\int x \, dx/(x^2-9)} = e^{\frac{1}{2} \int 2x \, dx/(x^2-9)} = e^{\frac{1}{2} \ln|x^2-9|} = \sqrt{x^2-9}.$$

After multiplying the standard form (7) by this factor, we get

$$\frac{d}{dx} \left[\sqrt{x^2-9} y \right] = 0.$$

Integrating both sides of the last equation gives $\sqrt{x^2-9} y = c$. Thus for either $x > 3$ or $x < -3$ the general solution of the equation is $y = \frac{c}{\sqrt{x^2-9}}$. ≡

Notice in Example 4 that $x = 3$ and $x = -3$ are singular points of the equation and that every function in the general solution $y = c/\sqrt{x^2-9}$ is discontinuous at these points. On the other hand, $x = 0$ is a singular point of the differential equation in Example 3, but the general solution $y = x^5e^x - x^4e^x + cx^4$ is noteworthy in that every function in this one-parameter family is continuous at $x = 0$ and is defined on the interval $(-\infty, \infty)$ and not just on $(0, \infty)$, as stated in the solution. However, the family $y = x^5e^x - x^4e^x + cx^4$ defined on $(-\infty, \infty)$ cannot be considered the general solution of the DE, since the singular point $x = 0$ still causes a problem. See Problems 45 and 46 in Exercises 2.3.

EXAMPLE 5 An Initial-Value Problem

Solve $\frac{dy}{dx} + y = x$, $y(0) = 4$.

SOLUTION The equation is in standard form, and $P(x) = 1$ and $f(x) = x$ are continuous on $(-\infty, \infty)$. The integrating factor is $e^{\int dx} = e^x$, so integrating

$$\frac{d}{dx} [e^x y] = xe^x$$

gives $e^x y = xe^x - e^x + c$. Solving this last equation for y yields the general solution $y = x - 1 + ce^{-x}$. But from the initial condition we know that $y = 4$ when $x = 0$. Substituting these values into the general solution implies that $c = 5$. Hence the solution of the problem is

$$y = x - 1 + 5e^{-x}, \quad -\infty < x < \infty. \quad (8) \quad \equiv$$

Figure 2.3.2, obtained with the aid of a graphing utility, shows the graph of the solution (8) in dark blue along with the graphs of other members of the one-parameter family of solutions $y = x - 1 + ce^{-x}$. It is interesting to observe that as x increases, the graphs of all members of this family are close to the graph of the solution $y = x - 1$. The last solution corresponds to $c = 0$ in the family and is shown in

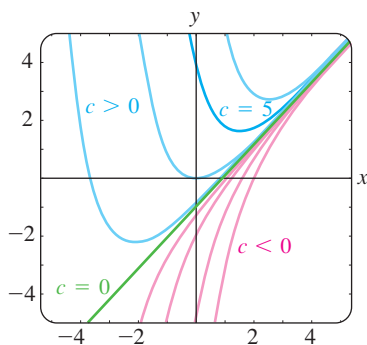


FIGURE 2.3.2 Solution curves of DE in Example 5

dark green in Figure 2.3.2. This asymptotic behavior of solutions is due to the fact that the contribution of ce^{-x} , $c \neq 0$, becomes negligible for increasing values of x . We say that ce^{-x} is a **transient term**, since $e^{-x} \rightarrow 0$ as $x \rightarrow \infty$. While this behavior is not characteristic of all general solutions of linear equations (see Example 2), the notion of a transient is often important in applied problems.

Discontinuous Coefficient In applications, the coefficients $P(x)$ and $f(x)$ in (2) may be piecewise continuous. In the next example $f(x)$ is piecewise continuous on $[0, \infty)$ with a single discontinuity, namely, a (finite) jump discontinuity at $x = 1$. We solve the problem in two parts corresponding to the two intervals over which f is defined. It is then possible to piece together the two solutions at $x = 1$ so that $y(x)$ is continuous on $[0, \infty)$.

EXAMPLE 6 An Initial-Value Problem

$$\text{Solve } \frac{dy}{dx} + y = f(x), \quad y(0) = 0 \quad \text{where} \quad f(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & x > 1. \end{cases}$$

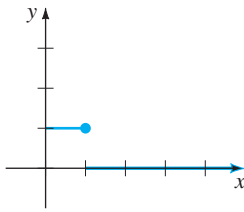


FIGURE 2.3.3 Discontinuous $f(x)$ in Example 6

SOLUTION The graph of the discontinuous function f is shown in Figure 2.3.3. We solve the DE for $y(x)$ first on the interval $[0, 1]$ and then on the interval $(1, \infty)$. For $0 \leq x \leq 1$ we have

$$\frac{dy}{dx} + y = 1 \quad \text{or, equivalently,} \quad \frac{d}{dx}[e^x y] = e^x.$$

Integrating this last equation and solving for y gives $y = 1 + c_1 e^{-x}$. Since $y(0) = 0$, we must have $c_1 = -1$, and therefore $y = 1 - e^{-x}$, $0 \leq x \leq 1$. Then for $x > 1$ the equation

$$\frac{dy}{dx} + y = 0$$

leads to $y = c_2 e^{-x}$. Hence we can write

$$y = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1, \\ c_2 e^{-x}, & x > 1. \end{cases}$$

By appealing to the definition of continuity at a point, it is possible to determine c_2 so that the foregoing function is continuous at $x = 1$. The requirement that $\lim_{x \rightarrow 1^+} y(x) = y(1)$ implies that $c_2 e^{-1} = 1 - e^{-1}$ or $c_2 = e - 1$. As seen in Figure 2.3.4, the function

$$y = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1, \\ (e - 1)e^{-x}, & x > 1 \end{cases} \quad (9)$$

is continuous on $(0, \infty)$.

It is worthwhile to think about (9) and Figure 2.3.4 a little bit; you are urged to read and answer Problem 48 in Exercises 2.3.

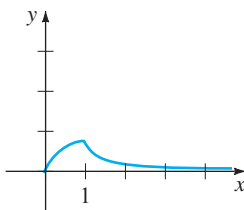


FIGURE 2.3.4 Graph of (9) in Example 6

Functions Defined by Integrals At the end of Section 2.2 we discussed the fact that some simple continuous functions do not possess antiderivatives that are elementary functions and that integrals of these kinds of functions are called **nonelementary**. For example, you may have seen in calculus that $\int e^{-x^2} dx$ and $\int \sin x^2 dx$ are nonelementary integrals. In applied mathematics some important functions are *defined* in terms of nonelementary integrals. Two such **special functions** are the **error function** and **complementary error function**:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad \text{and} \quad \operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt. \quad (10)$$

From the known result $\int_0^\infty e^{-t^2} dt = \sqrt{\pi}/2^*$ we can write $(2/\sqrt{\pi}) \int_0^\infty e^{-t^2} dt = 1$. Then from $\int_0^\infty = \int_0^x + \int_x^\infty$ it is seen from (10) that the complementary error function $\operatorname{erfc}(x)$ is related to $\operatorname{erf}(x)$ by $\operatorname{erf}(x) + \operatorname{erfc}(x) = 1$. Because of its importance in probability, statistics, and applied partial differential equations, the error function has been extensively tabulated. Note that $\operatorname{erf}(0) = 0$ is one obvious function value. Values of $\operatorname{erf}(x)$ can also be found by using a CAS.

EXAMPLE 7 The Error Function

Solve the initial-value problem $\frac{dy}{dx} - 2xy = 2, \quad y(0) = 1$.

SOLUTION Since the equation is already in standard form, we see that the integrating factor is e^{-x^2} , so from

$$\frac{d}{dx}[e^{-x^2}y] = 2e^{-x^2} \quad \text{we get} \quad y = 2e^{x^2} \int_0^x e^{-t^2} dt + ce^{x^2}. \quad (11)$$

Applying $y(0) = 1$ to the last expression then gives $c = 1$. Hence the solution of the problem is

$$y = 2e^{x^2} \int_0^x e^{-t^2} dt + e^{x^2} \quad \text{or} \quad y = e^{x^2}[1 + \sqrt{\pi} \operatorname{erf}(x)].$$

The graph of this solution on the interval $(-\infty, \infty)$, shown in dark blue in Figure 2.3.5 among other members of the family defined in (11), was obtained with the aid of a computer algebra system.

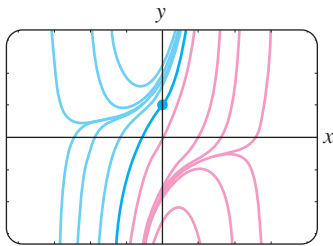


FIGURE 2.3.5 Solution curves of DE in Example 7

Use of Computers The computer algebra systems *Mathematica* and *Maple* are capable of producing implicit or explicit solutions for some kinds of differential equations using their *dsolve* commands.[†]

REMARKS

(i) A linear first-order differential equation

$$a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

is said to be **homogeneous**, whereas the equation

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

with $g(x)$ not identically zero is said to be **nonhomogeneous**. For example, the linear equations $xy' + y = 0$ and $xy' + y = e^x$ are, in turn, homogeneous and nonhomogeneous. As can be seen in this example the trivial solution $y = 0$ is always a solution of a homogeneous linear DE. Store this terminology in the back of your mind because it becomes important when we study linear higher-order ordinary differential equations in Chapter 4.

*This result is usually proved in the third semester of calculus.

[†]Certain commands have the same spelling, but in *Mathematica* commands begin with a capital letter (**DSolve**), whereas in *Maple* the same command begins with a lower case letter (**dsolve**). When discussing such common syntax, we compromise and write, for example, *dsolve*. See the *Student Resource Manual* for the complete input commands used to solve a linear first-order DE.

(ii) Occasionally, a first-order differential equation is not linear in one variable but is linear in the other variable. For example, the differential equation

$$\frac{dy}{dx} = \frac{1}{x + y^2}$$

is not linear in the variable y . But its reciprocal

$$\frac{dx}{dy} = x + y^2 \quad \text{or} \quad \frac{dx}{dy} - x = y^2$$

is recognized as linear in the variable x . You should verify that the integrating factor $e^{\int(-1)dy} = e^{-y}$ and integration by parts yield the explicit solution $x = -y^2 - 2y - 2 + ce^y$ for the second equation. This expression is then an implicit solution of the first equation

(iii) Mathematicians have adopted as their own certain words from engineering, which they found appropriately descriptive. The word *transient*, used earlier, is one of these terms. In future discussions the words *input* and *output* will occasionally pop up. The function f in (2) is called the **input** or **driving function**; a solution $y(x)$ of the differential equation for a given input is called the **output** or **response**.

(iv) The term **special functions** mentioned in conjunction with the error function also applies to the **sine integral function** and the **Fresnel sine integral** introduced in Problems 55 and 56 in Exercises 2.3. “Special Functions” is actually a well-defined branch of mathematics. More special functions are studied in Section 6.4.

EXERCISES 2.3

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–24 find the general solution of the given differential equation. Give the largest interval I over which the general solution is defined. Determine whether there are any transient terms in the general solution.

1. $\frac{dy}{dx} = 5y$
2. $\frac{dy}{dx} + 2y = 0$
3. $\frac{dy}{dx} + y = e^{3x}$
4. $3\frac{dy}{dx} + 12y = 4$
5. $y' + 3x^2y = x^2$
6. $y' + 2xy = x^3$
7. $x^2y' + xy = 1$
8. $y' = 2y + x^2 + 5$
9. $x\frac{dy}{dx} - y = x^2\sin x$
10. $x\frac{dy}{dx} + 2y = 3$
11. $x\frac{dy}{dx} + 4y = x^3 - x$
12. $(1+x)\frac{dy}{dx} - xy = x + x^2$
13. $x^2y' + x(x+2)y = e^x$
14. $xy' + (1+x)y = e^{-x}\sin 2x$
15. $y\,dx - 4(x+y^6)\,dy = 0$
16. $y\,dx = (ye^y - 2x)\,dy$
17. $\cos x\frac{dy}{dx} + (\sin x)y = 1$

$$18. \cos^2 x \sin x \frac{dy}{dx} + (\cos^3 x)y = 1$$

$$19. (x+1)\frac{dy}{dx} + (x+2)y = 2xe^{-x}$$

$$20. (x+2)^2\frac{dy}{dx} = 5 - 8y - 4xy$$

$$21. \frac{dr}{d\theta} + r \sec \theta = \cos \theta$$

$$22. \frac{dP}{dt} + 2tP = P + 4t - 2$$

$$23. x\frac{dy}{dx} + (3x+1)y = e^{-3x}$$

$$24. (x^2-1)\frac{dy}{dx} + 2y = (x+1)^2$$

In Problems 25–36 solve the given initial-value problem. Give the largest interval I over which the solution is defined

$$25. \frac{dy}{dx} = x + 5y, \quad y(0) = 3$$

$$26. \frac{dy}{dx} = 2x - 3y, \quad y(0) = \frac{1}{3}$$

$$27. xy' + y = e^x, \quad y(1) = 2$$

28. $y \frac{dx}{dy} - x = 2y^2, \quad y(1) = 5$
29. $L \frac{di}{dt} + Ri = E, \quad i(0) = i_0, \quad L, R, E, i_0 \text{ constants}$
30. $\frac{dT}{dt} = k(T - T_m), \quad T(0) = T_0, \quad k, T_m, T_0 \text{ constants}$
31. $x \frac{dy}{dx} + y = 4x + 1, \quad y(1) = 8$
32. $y' + 4xy = x^3 e^{x^2}, \quad y(0) = -1$
33. $(x + 1) \frac{dy}{dx} + y = \ln x, \quad y(1) = 10$
34. $x(x + 1) \frac{dy}{dx} + xy = 1, \quad y(e) = 1$
35. $y' - (\sin x)y = 2 \sin x, \quad y(\pi/2) = 1$
36. $y' + (\tan x)y = \cos^2 x, \quad y(0) = -1$

In Problems 37–40 proceed as in Example 6 to solve the given initial-value problem. Use a graphing utility to graph the continuous function $y(x)$.

37. $\frac{dy}{dx} + 2y = f(x), \quad y(0) = 0, \quad \text{where}$

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 3 \\ 0, & x > 3 \end{cases}$$

38. $\frac{dy}{dx} + y = f(x), \quad y(0) = 1, \quad \text{where}$

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ -1, & x > 1 \end{cases}$$

39. $\frac{dy}{dx} + 2xy = f(x), \quad y(0) = 2, \quad \text{where}$

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$$

40. $(1 + x^2) \frac{dy}{dx} + 2xy = f(x), \quad y(0) = 0, \quad \text{where}$

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ -x, & x \geq 1 \end{cases}$$

41. Proceed in a manner analogous to Example 6 to solve the initial-value problem $y' + P(x)y = 4x, \quad y(0) = 3$, where

$$P(x) = \begin{cases} 2, & 0 \leq x \leq 1, \\ -2/x, & x > 1. \end{cases}$$

Use a graphing utility to graph the continuous function $y(x)$.

42. Consider the initial-value problem $y' + e^x y = f(x), \quad y(0) = 1$. Express the solution of the IVP for $x > 0$ as a nonelementary integral when $f(x) = 1$. What is the solution when $f(x) = 0$? When $f(x) = e^x$?
43. Express the solution of the initial-value problem $y' - 2xy = 1, \quad y(1) = 1$, in terms of $\text{erf}(x)$.

Discussion Problems

44. Reread the discussion following Example 2. Construct a linear first-order differential equation for which all nonconstant solutions approach the horizontal asymptote $y = 4$ as $x \rightarrow \infty$.
45. Reread Example 3 and then discuss, with reference to Theorem 1.2.1, the existence and uniqueness of a solution of the initial-value problem consisting of $xy' - 4y = x^6 e^x$ and the given initial condition.
- (a) $y(0) = 0$ (b) $y(0) = y_0, y_0 > 0$
(c) $y(x_0) = y_0, x_0 > 0, y_0 > 0$
46. Reread Example 4 and then find the general solution of the differential equation on the interval $(-3, 3)$.
47. Reread the discussion following Example 5. Construct a linear first-order differential equation for which all solutions are asymptotic to the line $y = 3x - 5$ as $x \rightarrow \infty$.
48. Reread Example 6 and then discuss why it is technically incorrect to say that the function in (9) is a “solution” of the IVP on the interval $[0, \infty)$.
49. (a) Construct a linear first-order differential equation of the form $xy' + a_0(x)y = g(x)$ for which $y_c = c/x^3$ and $y_p = x^3$. Give an interval on which $y = x^3 + c/x^3$ is the general solution of the DE.
- (b) Give an initial condition $y(x_0) = y_0$ for the DE found in part (a) so that the solution of the IVP is $y = x^3 - 1/x^3$. Repeat if the solution is $y = x^3 + 2/x^3$. Give an interval I of definition of each of these solutions. Graph the solution curves. Is there an initial-value problem whose solution is defined on $-\infty, \infty$?
- (c) Is each IVP found in part (b) unique? That is, can there be more than one IVP for which, say, $y = x^3 - 1/x^3, x$ in some interval I , is the solution?
50. In determining the integrating factor (3), we did not use a constant of integration in the evaluation of $\int P(x) dx$. Explain why using $\int P(x) dx + c_1$ has no effect on the solution of (2).
51. Suppose $P(x)$ is continuous on some interval I and a is a number in I . What can be said about the solution of the initial-value problem $y' + P(x)y = 0, \quad y(a) = 0$?

Mathematical Models

52. **Radioactive Decay Series** The following system of differential equations is encountered in the study of the decay of a special type of radioactive series of elements:

$$\begin{aligned} \frac{dx}{dt} &= -\lambda_1 x \\ \frac{dy}{dt} &= \lambda_1 x - \lambda_2 y, \end{aligned}$$

where λ_1 and λ_2 are constants. Discuss how to solve this system subject to $x(0) = x_0, y(0) = y_0$. Carry out your ideas.